

Foundational Models for Meme Understanding

Evaluating Vision-Language Models on Hateful Meme Benchmarks

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Abstract

Memes combine visual content, embedded text, humour, irony, and cultural references, making them difficult for automatic content moderation systems to interpret. This project studies the ability of foundation vision-language models (VLMs) to detect harmful content in memes, with a focus on hateful and misogynistic content. We evaluated our approach on two complementary datasets: FHM-FG, which provides fine-grained annotations for hateful memes, and MAMI, which focuses on misogyny in multimodal memes. We compare generative VLMs, namely LLaVA 7B and Qwen 7B, with an encoder-based SigLIP/BERT pipeline. Our methodology investigates how multimodal input, OCR-extracted text, and prompt design influence classification performance. We report quantitative results using accuracy and Macro F1-score, and complement them with a qualitative analysis of model errors. Overall, our findings show that current models can identify explicit harmful content, but still struggle with implicit hate, cultural context, hallucinated visual or textual interpretations, and cases where the harmful meaning only emerges from the interaction between image and text.

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1 Introduction

Online platforms have become a primary medium for communication and information exchange, but they have also facilitated the spread of harmful content at an unprecedented scale. Among the most concerning forms of online harm is hate speech (HS). Although no universally accepted definition exists, HS is generally understood as content that attacks, degrades, excludes, threatens, or promotes hostility toward individuals or groups based on protected characteristics such as race, ethnicity, nationality, religion, gender, sexual orientation, or disability (Erjavec and Kovačič, 2012). Such content can range from explicit insults and slurs to more subtle forms of stereotyping, dehumanization, and incitement to violence.

The detection of harmful online content has become increasingly important due to its societal impact, and recent regulatory initiatives, such as the European Union’s Digital Services Act (DSA), require large online platforms to assess and mitigate risks associated with illegal and harmful content. As a result, automated moderation systems have become a major research focus.¹

Memes represent a particularly challenging form of harmful content. By combining images and text, memes often rely on humour, irony, cultural references, and contextual knowledge that make them difficult to interpret automatically. Unlike conventional image-text pairs, their meaning frequently emerges from the interaction between visual and textual information rather than from either modality alone, making harmful meme understanding a difficult open problem for current VLMs (Kiela et al., 2020; Shahroor et al., 2026).

This paper presents a systematic evaluation of current VLMs capabilities for the task of hateful meme detection. To evaluate these models, we build on the progressive annotation chain initiated by FHM (Kiela et al., 2020) and extended by FHM-FG (Mathias et al., 2021) and HatReD (detailed in Section 3.1), and complement it with the MAMI dataset (Fersini et al., 2022) for misogyny-specific content. Our main contributions are:

- A curated and balanced multi-label dataset derived from FHM-FG (enriched with HatReD reasoning) and MAMI.
- A two-stage evaluation: first, a comparison of prompting strategies (zero-shot, few-shot, and chain-of-thought) on LLaVA 7B and Qwen 7B; second, an evaluation using a fixed prompt for detailed quantitative and qualitative error analysis, alongside a baseline comparison against a SigLIP+BERT model.

2 Related Work

2.1 Algorithmic Approaches to Hateful Meme Detection

A number of previous works have studied hateful meme detection. The Hateful Memes Challenge introduced a widely adopted benchmark for multimodal hate detection and showed that unimodal models are insufficient, as hatefulness often emerges from the interaction between image and text rather than from either modality alone (Kiela et al., 2020). While early approaches relied on multimodal pipelines that combined visual features with OCR-extracted text, later work used pretrained multimodal models, such as VisualBERT with ensemble learning (Velioglu and Rose, 2020). More recent approaches rely on prompting and vision-language models, such as PromptHate (Cao et al., 2022), reflecting a shift from CNN/OCR-based fusion to pretrained multimodal and VLM-based methods. For an in-depth overview of datasets and methods for harmful meme detection, see Sharma et al. (2022).

2.2 VLMs

As highlighted in previous work, hateful meme detection requires models that can account for both visual and textual signals. This motivates the use of VLMs, which are designed to process image and text jointly and can therefore better capture multimodal cues such as humour, irony, stereotypes, and implicit harmful meaning (Afridi et al., 2020; Sharma et al., 2022; Nguyen and Ng, 2024).

¹digital-strategy.ec.europa.eu

Among generative VLMs, LLaVA uses visual instruction tuning (Liu et al., 2023), while Qwen-VL is designed for image-text understanding, localization, and visual reasoning (Bai et al., 2023). Both can process images and text and generate natural-language answers, making them suitable for prompt-based meme classification.

Encoder-based approaches also remain relevant: BERT encodes text (Devlin et al., 2019), while SigLIP provides visual-language representations through image-text pretraining with a sigmoid loss (Zhai et al., 2023). Together, these represent two main multimodal directions: generative VLMs such as LLaVA and Qwen-VL, and encoder-based pipelines such as SigLIP and BERT.

2.3 Prompt Engineering for Classification

Prompt engineering enables the use of LLMs and VLMs without costly weight fine-tuning. By structuring instructions and inputs, it helps align model behaviour with downstream classification goals. For multimodal meme understanding, the main prompting methods are zero-shot, few-shot, and Chain-of-Thought prompting.

Zero-Shot Prompting. Zero-shot prompting uses the pre-existing knowledge and visual-textual alignment learned by VLMs during pre-training. The model receives a task description, a target image, and optional OCR-extracted text, then predicts the label without prior examples.

Few-Shot Prompting. Few-shot prompting uses in-context learning (ICL) (Brown et al., 2020) by providing a few input-output examples, usually 2 to 5, directly in the prompt. This helps the model learn the expected output format, style, and decision boundaries.

Chain-of-Thought (CoT) Prompting. CoT prompting asks the model to generate intermediate reasoning steps before the final answer (Wei et al., 2022). For memes, this can help the model describe the image, read the text, analyse their interaction, and then select labels.

2.4 Datasets for Harmful Meme Detection

The growing prevalence of harmful memes on social media has motivated the development of several multimodal benchmarks for content moderation. The Facebook Hateful Memes (FHM) dataset (Kiela et al., 2020) introduced multimodal hate detection through image-text pairs requiring joint reasoning, but is limited to binary classification.

To address this limitation, Fine-Grained Hateful Memes (FHM-FG) (Mathias et al., 2021) extends FHM with annotations for targeted groups and attack types such as mocking, dehumanization, inferiority, exclusion, and incitement to violence. More recent datasets further broaden the scope of harmful content. MAMI (Fersini et al., 2022) focuses on misogyny through categories including shaming, stereotypes, objectification, and violence, while GuardHarMem (El Amrany et al., 2025) and MemeSafetyBench (Lee et al., 2024) cover a wider range of online harms.

Recent work has also emphasized multilingual and culturally diverse benchmarks. Bengali hateful meme datasets (Karim et al., 2022; Hossain et al., 2024) demonstrate the importance of language and cultural context, while Toxic Memes (Martinez et al., 2024) argues that binary labels are insufficient for capturing the mechanisms and intent underlying harmful content. Collectively, these benchmarks reflect a shift from binary detection toward fine-grained harmful meme understanding.

3 Methodology

3.1 Overall Pipeline

Figure 1 summarizes the proposed classification pipeline. Each meme first undergoes OCR and text preprocessing to produce a cleaned textual representation alongside the original image. A decoupled OCR strategy was adopted because preliminary experiments showed that LLaVA performs poorly on text extraction tasks (see Section 3.1.4), frequently ignoring or hallucinating meme text. Moreover, unlike FHM-FG, the MAMI dataset does not provide meme text as part of the annotations, making OCR a necessary preprocessing step. To ensure a fair com-

parison and isolate classification performance from OCR capability, the same extracted text was provided to all models.

The image, cleaned text, and classification prompt are then supplied to the target model. Following the prompt comparison described in Section 3.4, the definition-based prompts presented in Listings 4 and 5 were selected for the final experiments. Depending on the dataset and task configuration, the model outputs either `not_hateful` or a set of fine-grained harmful content categories. The predictions are subsequently parsed and evaluated using accuracy, F1-score, Macro-F1, and Geometric Mean F1.

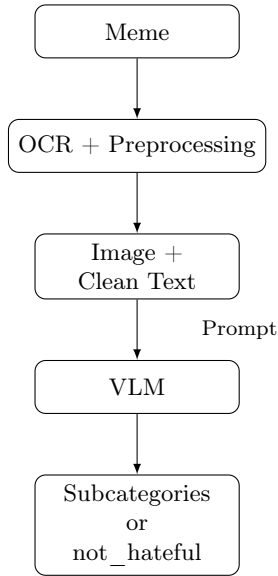


Figure 1: Overview of the harmful meme classification pipeline.

3.1.1 Dataset Selection and Justification

The datasets used in this work were selected to evaluate both general harmful meme understanding and domain-specific harmful content. **FHM-FG** was chosen because it extends the widely adopted **FHM** benchmark with fine-grained attack-type annotations, allowing evaluation beyond binary hate detection. **MAMI** was selected as a complementary benchmark focused on misogyny, providing a distinct set of harmful content categories and enabling the assessment of model behavior across different annotation taxonomies.

During the initial exploration phase, additional datasets such as **MemeSafetyBench**

were considered. However, their broader safety-oriented label space and annotation structure were less aligned with the multi-label classification setting targeted in this work. Consequently, the final evaluation focuses on **FHM-FG** and **MAMI**, which together provide complementary perspectives on fine-grained harmful meme understanding.

Figure 2 illustrates representative examples of the **MAMI** fine-grained categories.



Figure 2: Examples of misogynous memes.

3.1.2 Data Preprocessing Pipeline

1. **Text Preprocessing:** Text transcriptions are cleaned by removing stop words using NLTK, and tokens are subsequently lemmatised using spaCy (`en_core_web_sm`) to standardize textual inputs.
2. **Algorithmic Balancing:** To counter severe class imbalances, an algorithmic downsampling mechanism was implemented to ensure a uniform distribution across all multi-label categories.
3. **Binary Alignment:** An equal number

of non-hateful memes was dynamically selected to match the hateful subset, resulting in a perfectly balanced 50/50 binary classification split for a reliable evaluation.

3.1.3 Dataset Statistics

Following our preprocessing and balancing pipeline, Table 1 and Table 2 details the exact item distribution across our train and test splits for both evaluated sub-tasks.

Table 1: Class and Sub-category Distribution in the Cleaned FHM Dataset

Target Category	Train Split	Test Split	Total Items
Sub-task A (Binary)			
Hateful	320	80	400
Non-Hateful	320	80	400
Sub-task B (Multi-label)			
Race	64	16	80
Religion	64	16	80
Disability	64	16	80
Nationality	64	16	80
Sex	64	16	80
Total Dataset Size	640	120	800

Table 2: Class and Sub-category Distribution in the Cleaned MAMI Dataset

Target Category	Train Split	Test Split	Total Items
Sub-task A (Binary)			
Misogynous	288	72	360
Non-Misogynous	288	72	360
Sub-task B (Multi-label)			
Shaming	72	18	90
Stereotype	72	18	90
Objectification	72	18	90
Violence	72	18	90
Total Dataset Size	576	144	720

3.1.4 OCR Evaluation

Before harmful meme classification, we evaluated several OCR systems on FHM, which provides ground-truth meme text and therefore enables quantitative OCR evaluation. We compared traditional OCR tools (Tesseract² and EasyOCR³) with VLM-based approaches including DeepSeek OCR⁴, DeepSeek-VL2 Tiny⁵, Qwen2.5-VL-7B⁶ and

²<https://github.com/tesseract-ocr/tesseract>

³<https://github.com/JaidedAI/EasyOCR>

⁴<https://huggingface.co/deepseek-ai/DeepSeek-OCR>

⁵<https://huggingface.co/deepseek-ai/deepseek-vl2-tiny>

⁶<https://huggingface.co/Qwen/Qwen2.5-VL-7B-Instruct>

LLaVA-1.5-7B/13B.⁷

OCR quality was assessed using:

- Exact Match (EM), which measures whether the predicted text exactly matches the ground truth text;
- Smart EM, which allows for minor formatting differences such as punctuation or casing;
- Word EM, which evaluates correctness at the word level;
- Character Error Rate (CER), which computes the normalized edit distance at the character level; and
- Word Error Rate (WER), which measures the proportion of word-level errors.

As shown in Table 3, Qwen2.5-VL-7B achieved the highest exact-match performance (83%), while DeepSeek-VL2 Tiny obtained the lowest CER (0.013) and WER (0.022). In contrast, both LLaVA variants performed poorly, exhibiting frequent omissions and hallucinations. Given its strong accuracy and lower computational cost, DeepSeek-VL2 Tiny was selected as the OCR component for subsequent experiments.

Table 3: OCR extraction performance on the FHM dataset. Higher is better for EM), Smart EM, and Word EM, while lower is better for CER and WER.

Model	EM ↑	CER ↓	Smart EM ↑	Word EM ↑	WER ↓
LLaVA-1.5-7B	1%	0.836	5%	15%	0.920
LLaVA-1.5-13B	1%	0.956	6%	22%	1.102
Qwen2.5-VL-7B	83%	0.035	93%	91%	0.034
EasyOCR	5%	0.186	14%	5%	0.449
DeepSeek Tiny	79%	0.013	90%	86%	0.022
DeepSeek OCR	59%	0.049	75%	75%	0.061
Tesseract	0%	0.691	0%	1%	0.890

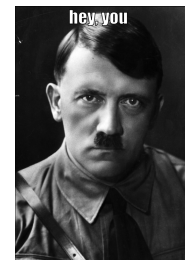


Figure 3: Example of an OCR failure produced by LLaVA-1.5-7B.

⁷<https://huggingface.co/llava-hf>

Figure 3 illustrates a representative OCR failure produced by LLaVA-1.5-7B. Rather than extracting the text “hey, you”, the model generates a generic description of the image content:

“A man with a mustache and a goatee is staring at the camera. He is wearing a brown shirt and a black tie. The man is also wearing a brown vest.”

This behavior was frequently observed during evaluation and suggests that LLaVA often defaults to image captioning instead of performing reliable text recognition, which helps explain its poor OCR performance in Table 3.

3.2 Hardware Setup

Experiments were conducted across three computing environments. Most VLM inference was performed either on the Télécom Paris computing cluster using NVIDIA Tesla P100 GPUs (16 GB VRAM) managed through SLURM, or on Google Colab instances equipped with NVIDIA Tesla T4 GPUs when cluster resources were unavailable or queue times became prohibitive. The supervised SigLIP + BERT baseline was trained locally on an NVIDIA RTX 2070 GPU (8 GB VRAM). All VLM experiments were executed in FP16 precision, while the supervised baseline was trained in FP32.

3.3 Models

Using the hardware infrastructure described in Section 3.2, we evaluated three model families: two generative VLMs, Qwen2.5-VL-7B and LLaVA-1.5-7B, and a supervised multimodal baseline combining SigLIP and BERT. Qwen and LLaVA were selected because they are widely adopted open-source VLMs with strong multimodal reasoning capabilities and have been used in recent studies on meme understanding and hateful meme analysis, while remaining small enough to run on the available academic computing infrastructure.

Qwen2.5-VL-7B. Qwen2.5-VL-7B-Instruct was used for zero-shot harmful meme classification. Inference was performed in FP16

without quantization, and the model was evaluated on the FHM-FG and MAMI test sets using the prompt-based classification framework described in Section 3.4.

LLaVA-1.5-7B. LLaVA-1.5-7B served as a second zero-shot VLM baseline. Similar to Qwen, it was evaluated using FP16 inference and the same prompt-based classification framework on both datasets.

SigLIP + BERT Fusion Baseline. As a supervised baseline, we used a multimodal fusion model combining a frozen SigLIP visual encoder (Zhai et al., 2023)⁸ and a frozen BERT text encoder (Devlin et al., 2019)⁹. Their embeddings were concatenated and passed through a lightweight MLP with binary and multi-label classification heads. The model was trained for 10 epochs and evaluated on both FHM-FG and MAMI.

Table 4: Characteristics of the evaluated models.

Model	Params	Year	Organization	Setting
Qwen2.5-VL-7B	7B	2024	Alibaba	Zero-shot
LLaVA-1.5-7B	7B	2023	UW/MSR	Zero-shot
SigLIP+BERT	~196M	2023/18	Google	Supervised

3.4 Prompt Engineering

For FHM-FG, we first compared three prompting strategies, zero-shot, few-shot, and CoT, on both LLaVA-1.5-7B and Qwen2.5-VL-7B, using the preliminary prompt templates reported in Appendix A.1. The results are shown in Table 5. Zero-shot prompting gave the best overall results and the most stable parseable outputs, while few-shot prompting sometimes caused copying from the examples and CoT often produced long responses that were harder to parse automatically. We therefore used zero-shot prompting for the final quantitative experiments.

Within the zero-shot setting, we tested a *Custom* prompt, reported in Appendix A.2, and two EU-based variants, reported in Appendix A.4. The Custom prompt used

⁸<https://huggingface.co/google/siglip-base-patch16-224>

⁹<https://huggingface.co/google-bert/bert-base-uncased>

Table 5: Preliminary comparison of prompting strategies on FHM-FG.

Model	Prompt Type	F1
LLaVA-1.5-7B	Zero-shot	0.69
LLaVA-1.5-7B	Few-shot	0.59
LLaVA-1.5-7B	CoT	0.62
Qwen2.5-VL-7B	Zero-shot	0.80
Qwen2.5-VL-7B	Few-shot	0.74
Qwen2.5-VL-7B	CoT	0.00

dataset-specific definitions, whereas the EU prompt reformulated the task using broader European Union hate-speech moderation principles. The EU+Reason variant additionally asked for a short explanation after the labels. The Custom setting was retained for the main experiments because it was more directly aligned with the FHM-FG taxonomy and gave better results.

Table 6: Zero-shot prompt comparison on FHM-FG.

Prompt Variant	F1
Custom definitions	0.6571
EU definitions	0.6190
EU with explanation	0.6087

For MAMI, we directly evaluated zero-shot prompt variants, assuming that the prompting strategy selected on FHM-FG would behave similarly on the second dataset. We tested a *Custom* prompt based directly on the dataset categories and a broader *Research* prompt based on definitions of misogyny and gender-based online abuse, both reported in Appendix A.3. The Standard setting was retained because it was more directly aligned with the MAMI taxonomy and gave better results.

Table 7: Zero-shot prompt comparison on MAMI.

Prompt Variant	F1
Custom definitions	0.7600
Research-based	0.5128

3.5 Evaluation Metrics

We report accuracy (A) and F1-score (F1) for binary classification, and per-class F1, Macro-F1, and Geometric Mean F1 (GMean-F1) for

multi-label classification. Accuracy measures overall correctness, but it can be misleading in our setting because some fine-grained categories are less represented or easier to miss; a model can therefore obtain a high accuracy while failing to detect specific harmful labels. F1-score is more informative because it balances precision and recall, capturing both false positives and false negatives. For the fine-grained labels, macro-F1 gives equal weight to all categories, while geometric mean F1 penalizes models that completely miss one category.

4 Results

4.1 Quantitative Results

The quantitative evaluation was conducted on two datasets: FHM-FG and MAMI. Experiments were performed on balanced test subsets of 160 and 144 samples, respectively. For each dataset, models were evaluated under two settings: binary classification, where the objective is to determine whether a meme is harmful (or misogynistic) or not, and fine-grained multi-label classification, where the goal is to identify the specific categories of harm present in the meme. Results are reported using A and F1 for binary classification, and per-class F1, Macro-F1, and GMean-F1 for fine-grained classification.

4.1.1 FHM-FG Results

Table 8: Binary classification results on FHM-FG.

Model	A	F1
LLaVA-1.5-7B	0.5750	0.3061
Qwen2.5-VL-7B	0.7063	0.6619
SigLIP+BERT fusion	0.6625	0.6707

Table 9: Aggregate fine-grained F1 results on FHM-FG.

Model	Macro-F1	GMean-F1
LLaVA-1.5-7B	0.2182	0.1933
Qwen2.5-VL-7B	0.2398	0.0457
SigLIP+BERT fusion	0.3186	0.2871

Table 10: Per-class F1-scores on FHM-FG attack types, part 1.

Model	Contempt	Inferiority	Dehuman.	Violence
LLaVA-1.5-7B	0.3125	0.1852	0.3103	0.2941
Qwen2.5-VL-7B	0.3226	0.0500	0.2083	0.2105
SigLIP+BERT fusion	0.2326	0.3731	0.4061	0.3014

Table 11: Per-class F1-scores on FHM-FG attack types, part 2.

Model	Mocking	Slurs	Exclusion
LLaVA-1.5-7B	0.1667	0.0870	0.0833
Qwen2.5-VL-7B	0.1176	0.3478	0.0000
SigLIP+BERT fusion	0.2025	0.1519	0.2105

On FHM-FG, Qwen2.5-VL-7B obtains the highest A, while SigLIP + BERT fusion obtains the highest F1 and the strongest aggregate fine-grained results. The per-class F1-scores show that fine-grained attack detection remains difficult.

4.1.2 MAMI Results

Table 12: Binary classification results on MAMI.

Model	A	F1
LLaVA-1.5-7B	0.6528	0.6212
Qwen2.5-VL-7B	0.6806	0.7356
SigLIP+BERT fusion	0.7986	0.7883

Table 13: Aggregate fine-grained F1 results on MAMI.

Model	Macro-F1	GMean-F1
LLaVA-1.5-7B	0.4472	0.4268
Qwen2.5-VL-7B	0.5952	0.5904
SigLIP+BERT fusion	0.4779	0.4490

Table 14: Per-class F1-scores on MAMI categories.

Model	Shaming	Stereotype	Objectif.	Violence
LLaVA-1.5-7B	0.2368	0.5000	0.4524	0.4255
Qwen2.5-VL-7B	0.5000	0.6000	0.5778	0.5625
SigLIP+BERT fusion	0.3111	0.4405	0.5324	0.3175

On MAMI, SigLIP + BERT fusion obtains the best binary accuracy and binary F1-score. However, Qwen2.5-VL-7B obtains the best aggregate fine-grained performance, with the highest

macro mean F1 and geometric mean F1. It also performs best across all four MAMI fine-grained categories. This suggests that Qwen2.5-VL-7B is more effective for identifying misogyny subtypes, while SigLIP + BERT fusion is stronger for the overall binary misogyny detection task.

4.2 Qualitative Comparison with HatReD Reasoning (Qwen 7B)

To assess whether Qwen2.5-VL-7B’s free-text explanations capture the same reasoning as human annotators, we compared its generated justifications against the ground-truth reasoning provided by HatReD on a 50-sample subset of FHM-FG. Of the 23 truly hateful memes in this subset, all were successfully matched to a HatReD reasoning annotation via their meme identifier, enabling a direct qualitative comparison. Qwen failed to flag 10 of these 23 memes (43%) as hateful at all, typically dismissing them as “humorous” or “non-offensive” when relying on the textual cue alone, suggesting that the model under-uses the visual modality when the meme text is ambiguous in isolation. Among the 13 memes correctly flagged as hateful, the model’s reasoning named the same targeted group as HatReD in 8 cases (62%), while the remaining 5 relied on generic phrasing such as “a group” or “the targeted group” without identifying who was actually targeted.

Beyond this aggregate pattern, the comparison reveals two recurring qualitative failure modes. First, Qwen tends to capture the surface tone of a meme (e.g., humour, mockery) without identifying the underlying dehumanising mechanism: for the meme “bride wear white? match dishwasher, stove fridge”, the model correctly labelled the content as *mocking* but framed it as disrespect toward a wedding tradition, whereas HatReD’s reasoning explains that the meme “dehumanizes the females as less capable humans suited for household chores”. Second, the model can produce a reasoning trace that is descriptively accurate yet inconsistent with its own final label: for “ever notice christians horrible drivers?”, Qwen’s explanation correctly identifies the mocking of Christians’ driving skills — closely matching HatReD’s own reasoning — yet still outputs the label *not_hateful* alongside *mocking*. Conversely, on

more explicit cases involving direct incitement or slurs (ids 07254 and 85096), **Qwen**’s reasoning converges closely with **HatReD**’s, correctly identifying both the targeted group and the specific harmful mechanism (incitement to violence, use of a slur). This suggests that the model’s natural-language explanations are most reliable for explicit harm and degrade for memes whose hatefulness emerges from implicit, culturally-coded dehumanisation.



(a) id 59876



(b) id 74021



(c) id 07254



(d) id 85096

Figure 4: The four **FHM-FG** memes discussed in the qualitative comparison with **HatReD** reasoning.

4.3 Comparison Between **FHM-FG** and **MAMI**

Overall, the models perform better on **MAMI** than on **FHM-FG**. This can be partly explained by the dataset structure. Its fine-grained labels, such as *shaming*, *stereotype*, *objectification*, and *violence*, are also more clearly separated and easier to interpret.

By contrast, **FHM-FG** is more challenging because its attack-type labels are subtler and sometimes semantically close. Distinguishing between categories such as *mocking* and *contempt* can be difficult even for human annotators, especially when memes rely on irony, implicit cultural references, or ambiguous humour. The stronger results on **MAMI** therefore do not necessarily mean that the task is easier in general, but rather that its labels are more balanced and distinguishable in our experimental setting.

4.4 Overall Model Performance

Overall, the models show complementary strengths. **Qwen2.5-VL-7B** performs best on fine-grained subcategory detection, especially on **MAMI**, suggesting that it handles detailed label distinctions better. However, on **FHM-FG**, its geometric mean F1 drops to 0.0457 because it never predicts the *exclusion* category correctly, giving that class an F1-score of 0.0. Since the geometric mean strongly penalizes any zero score, this greatly reduces the final metric despite reasonable performance on other categories.

LLaVA-1.5-7B generally performs worse, especially in terms of F1-score, indicating less stable zero-shot classification. **SigLIP + BERT** fusion performs best for binary classification on both datasets, likely because its encoder-based representations are better suited for separating hateful from non-hateful content, while generative VLMs such as **Qwen** appear stronger for detailed semantic categories.

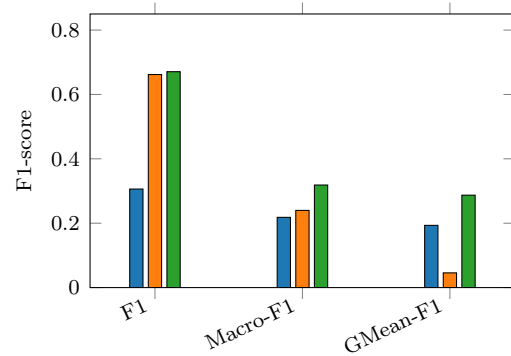


Figure 5: Model comparison on **FHM-FG** using binary F1, macro mean F1, and geometric mean F1.

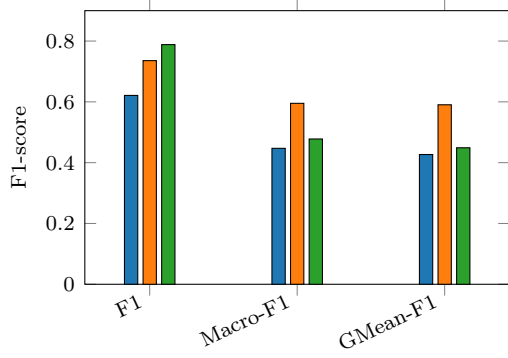


Figure 6: Model comparison on MAMI using F1, Macro-F1, and GMean-F1.

■ LLaVA-1.5-7B ■ Qwen2.5-VL-7B ■ SigLIP+BERT

5 Qualitative Error Analysis

5.1 Error Analysis on FHM-FG



(a) GT: Non-hateful.
LLaVA: Hateful. Qwen: Hateful.



(b) GT: Hateful.
LLaVA: Non-hateful.
Qwen: Hateful.



(c) GT: Non-hateful.
LLaVA: Non-hateful.
Qwen: Hateful.



(d) GT: Hateful.
LLaVA: Non-hateful.
Qwen: Non-hateful.

Figure 7: Representative failure cases from the FHM test set.

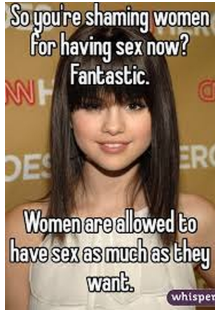
Figure 7 presents representative error cases from the FHM test set. Overall, LLaVA produced 68 errors (3 false positives, 65 false negatives), while Qwen produced 47 errors (13 false posi-

tives, 34 false negatives), indicating that both models are more likely to miss hateful content than to hallucinate it.

Example (a) illustrates a tendency of both models to over-rely on offensive lexical cues. The meme plays on the well-known expression “*I’m not racist, some of my best friends are black*” by replacing *friends* with *slaves*. Although offensive, it was annotated as non-hateful, yet both models classified it as hateful, suggesting difficulty distinguishing provocative racial humour from genuine hate speech. In contrast, Example (c) highlights Qwen’s tendency to over-trigger on politically charged language: although the meme was annotated as non-hateful, Qwen predicted a hateful label while LLaVA classified it correctly.

Example (d) demonstrates the importance of cultural and multimodal grounding. Correct interpretation requires combining the phrase “*sea monkeys*”, the depicted subjects, and the historical use of monkey-related comparisons as racist stereotypes. Both models failed to identify the hateful intent, revealing limitations in contextual reasoning. Finally, Example (b) illustrates a common false-negative pattern for LLaVA. The meme contains an explicit derogatory statement targeting a protected group, which Qwen correctly identified as hateful while LLaVA classified as non-hateful. Together, these examples show that current VLMs remain vulnerable to satire misunderstanding, political over-triggering, weak cultural grounding, and incomplete integration of visual and textual information.

5.2 Error Analysis on MAMI



(a) GT: Not misogynous.
LLaVA: stereotype, shaming.
Qwen: shaming.



(b) GT: violence. LLaVA: not misogynous.
Qwen: not misogynous.



(c) GT: stereotype, objectification. LLaVA: stereotype. Qwen: stereotype.



(d) GT: stereotype. LLaVA: not misogynous.
Qwen: stereotype.

Figure 8: Representative MAMI examples used for error analysis.

Figure 8 presents representative error cases from the MAMI dataset. Overall, LLaVA produces 50 errors (31 false negatives, 19 false positives), while Qwen produces 46 errors (8 false negatives, 38 false positives). Compared with FHM-FG, both models exhibit substantially higher false-positive rates, with Qwen showing a particularly strong tendency to hallucinate misogynistic content. Example (a) illustrates this behaviour: although the meme is annotated as non-misogynous and is in fact supportive of women, both models predict misogynistic categories. This reflects a broader pattern in which supportive or pro-women content is misinterpreted as shaming or misogyny due to the presence of gender-related cues.

Example (b) demonstrates a multimodal reasoning failure. Although annotated as *violence*, both models favour the benign poker interpretation of the phrase “beat my wife” and

fail to integrate the visual context reinforcing the violent meaning. Example (c) highlights difficulties in fine-grained classification: both models correctly identify a *stereotype* but fail to detect *objectification*, suggesting that deeper forms of misogyny remain challenging to capture.

Finally, Example (d) illustrates a common false-negative pattern for LLaVA. While Qwen correctly identifies the underlying stereotype, LLaVA classifies the meme as non-misogynous, indicating difficulty in recognizing implicit cultural stereotypes. Overall, these examples reveal limitations in stance detection, multimodal reasoning, and fine-grained misogyny understanding, while highlighting Qwen’s tendency toward over-prediction and LLaVA’s tendency toward under-detection.

6 Limitations

- **Dataset size:** Experiments were conducted on balanced subsets of approximately 800 samples for FHM-FG and 720 samples for MAMI. While suitable for comparative evaluation, these subsets may not fully capture the diversity of real-world harmful memes.
- **Cultural and linguistic coverage:** The datasets are predominantly English-language and Western-centric. As many memes rely on culture-specific references, stereotypes, and implicit knowledge, the results may not generalize well to other languages or communities.
- **OCR dependency:** Meme understanding often depends on accurate text extraction. stylised fonts, image compression, and complex backgrounds can introduce OCR errors that propagate to downstream classification.
- **Annotation subjectivity:** Distinguishing between humour, satire, offensiveness, and genuine harm is inherently subjective, particularly for implicit or context-dependent memes.
- **Computational constraints:** Experiments were conducted as specified in Section 3.2. Hardware limitations,

job queuing delays, and dependency-management issues restricted the number of models, hyperparameter configurations, and fine-tuning experiments that could be explored.

- **Model accessibility:** The study focused exclusively on open-source models to ensure reproducibility and low deployment costs. Although preliminary experiments suggested that proprietary systems such as GPT-5.4 may achieve higher performance, they were excluded from the main evaluation due to cost and accessibility considerations.

7 Challenges and Findings

7.1 Data team

The primary challenge was expanding our scope by finding other datasets to complement FHM. Misogyny was not our initial objective; we first explored general hate and sentiment datasets, preprocessing Memotion (Sharma et al., 2020) before pivoting due to its lack of explicit HS annotations. Securing an alternative was difficult, as datasets with harmful content require formal access requests.

7.2 Research team

A major challenge was the computational setup. The university cluster GPUs were relatively old for running large VLMs, which made inference slow and created several dependency issues. These constraints also prevented fine-tuning, since the available GPU memory was not sufficient.

Interpreting the results was also difficult because errors could come from many sources: OCR mistakes, VLM randomness, prompt sensitivity, meme subjectivity, or weak multimodal reasoning. This made it hard to identify the exact cause of each failure.

Overall, we found that the traditional SigLIP + BERT baseline remains stronger for binary meme classification, while VLMs such as Qwen are more effective for fine-grained subcategory prediction. This suggests that VLM accuracy could be further improved with fine-tuning on task-specific meme datasets.

8 Alignment with the Proposed Plan

8.1 Plan vs. Outcome

Overall, we managed to complete the main objectives of the project: selecting relevant datasets, preprocessing the data, testing several VLMs, evaluating their performance, and analysing their errors. However, the project did not fully follow the initial schedule, mainly because the report writing started later than expected.

Another difference from the initial plan was the OCR component. At first, OCR was not a major part of the project, but it became necessary because meme text extraction strongly affected model performance. Testing different OCR tools and integrating the OCR pipeline took a significant amount of time, but it also improved the quality and reliability of our experiments.

8.2 Lessons Learned

If we had to restart the project, we would use Google Colab more for early model testing, since it was faster to set up and had fewer dependency issues than the university cluster.

This project taught us how to run experiments on the cluster, manage GPU constraints, use large VLMs such as LLaVA and Qwen, and build an OCR pipeline. We also learned that meme classification is highly subjective and difficult, because interpretation often depends on humour, irony, cultural context, and the interaction between image and text.

9 Conclusion & Future Work

This work evaluated the effectiveness of modern VLMs for fine-grained harmful meme understanding on the FHM-FG and MAMI datasets. Through a comparison of Qwen2.5-VL-7B, LLaVA-1.5-7B, and a supervised SigLIP + BERT baseline, we found that Qwen generally achieved the strongest zero-shot performance, while the supervised baseline remained competitive despite its relatively small number of trainable parameters. Our OCR experiments further highlighted the importance of reliable

text extraction, motivating the use of external OCR and multimodal inputs combining images with extracted text. Error analysis revealed that all models continue to struggle with satire, implicit harmful content, cultural references, and fine-grained category distinctions, demonstrating that harmful meme understanding remains substantially more challenging than binary harmful/non-harmful classification. These findings emphasize both the promise of foundation models for content moderation and the need for improved multimodal reasoning and contextual understanding in future systems.

Future work.

- **Multilingual extension:** incorporating non-English meme datasets (e.g., Hindi-English code-mixed MIMIC dataset) to address cultural bias.
- **In-context learning:** providing richer contextual examples to anchor model reasoning.
- **Fine-tuning:** adapting LLaVA or Qwen on the curated FHM-FG dataset for improved performance.
- **Larger qualitative analysis:** extending the comparison with HatReD reasoning annotations to a larger subset of memes, in order to better evaluate whether model explanations match human reasoning.

10 Teams and Contributions

The six members split into two sub-teams working in parallel, coordinated through a WhatsApp group and weekly meetings with our supervisors. The **Data team** (Hugo FOULON, Philibert PIERSON DE BRABOIS, Théo RENAUDO) sourced, analysed, cleaned, and balanced the benchmarks; the **Research team** (Tia HAKMEH, Samir ALSABBAGH, Camil DAOU) handled model setup, prompting, OCR experiments, and evaluation. Individual contributions are as follows:

- **Hugo FOULON** — initial data exploration; preprocessed Meme Safety Bench

with an EasyOCR pipeline, then led the pivot to MAMI and designed its balancing pipeline.

- **Philibert PIERSON DE BRABOIS** — FHM-FG data engineering: the balanced 800-meme subset, the text-cleaning and target-isolation strategy, HurtLex-based semantic profiling, and the conversational JSON format for LLaVA.
- **Théo RENAUDO** — FHM and FHM-FG processing and HatReD reasoning enrichment; qualitative comparison of Qwen explanations against HatReD human annotations.
- **Camil DAOU** — model and OCR installation and cluster-environment management; training/inference pipelines, the supervised baseline, and evaluation scripts.
- **Samir ALSABBAGH** — design and systematic evaluation of zero-shot, few-shot, and CoT prompts (best of 30) on LLaVA and Qwen, and result interpretation.
- **Tia HAKMEH** — ran LLaVA and Qwen on FHM-FG and MAMI, computed the evaluation metrics, and led the error and failure-case analysis.

11 Use of AI Tools

11.1 In the Code

AI tools were used during the implementation process as support tools. Gemini 3.5 Flash was mainly used to research and compare possible prompt structures, especially for zero-shot and Chain-of-Thought prompting. GitHub Copilot with Claude Sonnet 4.6 was used to support coding tasks, such as debugging, understanding errors, improving code structure, and writing small Python snippets. GPT-5.5 was also used to clarify implementation issues and help debug some problems encountered during model execution.

However, the main implementation choices, experimental setup, dataset preprocessing, model execution, and evaluation logic were designed and controlled by the team. AI-generated suggestions were tested, corrected,

and adapted manually before being integrated into the final code.

11.2 In the Report

AI tools were used during the report writing process mainly to improve formulations, correct grammar, make the text more concise, and help respect the word limit. The main ideas, experiments, results, and interpretations came from our own work, while AI tools helped us express them more clearly and professionally.

The tools used included Gemini 3.5 Flash and GitHub Copilot with Claude Sonnet 4.6. They were also used to help write and format parts of the report in LaTeX, especially for tables, figures, captions, and paragraph restructuring. All AI-assisted content was reviewed, corrected, and adapted by the team before being included in the final report.

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A Prompt Templates

This appendix provides the full text of the prompt templates evaluated in this project. These include the three foundational strategies selected as the best of 30 candidates (10 of each

strategy), the refined zero-shot prompts used for final quantitative testing on FHM-FG and MAMI, and the experimental variants used to assess the impact of European Union (EU) legal principles and research-based misogyny definitions.

A.1 Top-Selected Preliminary Prompts

The following templates were identified as the most effective during the initial research phase on the FHM-FG dataset.

Listing 1: Zero-Shot Base Prompt

```
Act as an expert Trust and Safety
content moderator. Your task is
to classify the provided meme
based on the text and the image.
The text extracted from the
meme is: "{clean_text}".

You must assign the meme to one or
more of these categories based
on their definitions:
- contempt: Expressing extreme
  disdain or disgust.
- inferiority: Implying a group is
  less intelligent, capable, or
  human.
- dehumanising: Comparing people to
  animals, objects, insects, or
  diseases.
- inciting_violence: Encouraging,
  threatening, or celebrating
  physical harm.
- mocking: Cruelly making fun of a
  group's traits, tragedies, or
  stereotypes.
- slurs: Using derogatory language
  or hate speech terms.
- exclusion: Stating that a group
  does not belong or should be
  deported/removed.
- not_hateful: The meme is benign,
  or just a non-hateful joke.

YOUR OUTPUT MUST BE ONLY A JSON
ARRAY OF STRINGS representing
all applicable labels. Do not
include any other text or
markdown. Expected Output format
: ["dehumanising", "inferiority"]
or ["not_hateful"]
```

Listing 2: Multimodal Few-Shot Prompt

```
Classify the meme into one or more
of these labels: [contempt,
inferiority, dehumanising,
inciting_violence, mocking,
slurs, exclusion, not_hateful],
based on their definitions: [
Definitions truncated for
brevity]
```

--- EXAMPLES ---

Example 1: Text: "happy cinco de
ported!" | Visual: Man running
across a border. Answer: ["
exclusion", "mocking"]

Example 2: Text: "send me sexy pic"
| Visual: A normal portrait.
Answer: ["not_hateful"]

Example 3: Text: "we kill you on
9/11 and you elect 3 of us to
congress stupid infidels!" |
Visual: Political figures.
Answer: ["dehumanising", "
inciting_violence"]

--- TARGET MEME ---

Target Text: "{clean_text}"

Look at the provided image and read
the target text. Output ONLY
the JSON array for this target
meme. Do not add any
conversational text.

Listing 3: Structured Chain-of-Thought (CoT) Prompt

```
analyse this meme step-by-step to
determine if it contains hateful
attacks.
```

Step 1. Describe the visual
elements of the image
objectively.

Step 2. Read the extracted text: "{
clean_text}".

Step 3. Explain the juxtaposition:
How does the text change the
meaning of the image? Does the
combination create a malicious
attack against a specific group?

Step 4. Select ALL categories that
apply from this list: [contempt,
inferiority, dehumanising,
inciting_violence, mocking,
slurs, exclusion, not_hateful],
based on their definitions: [
Definitions truncated]

FINAL OUTPUT: On a new line at the

```

very end of your response,
output the final labels as a
JSON array starting with "LABELS
: ". Example format: LABELS: ["
contempt", "slurs"]

```

A.2 Final FHM-FG Testing Prompts

The following prompt was selected for the final quantitative benchmark on the FHM-FG dataset.

Listing 4: Custom Definitions Prompt (FHM-FG Final Test)

```

Act as an expert Trust and Safety
content moderator.
analyse both the image and the
accompanying meme text. The text
extracted from the meme is: "{
text}"
Consider the meaning created by the
combination of the image and
the text. Do not classify based
on the image or text alone.
Assign one or more of the following
labels according to these
definitions:
* contempt: Expressing extreme
disdain, disgust, hatred, or
hostility toward a group.
* inferiority: Suggesting that a
group is less intelligent, less
capable, less civilized, or less
human than others.
* dehumanising: Comparing people to
animals, insects, parasites,
objects, diseases, or other non-
human entities.
* inciting_violence: Encouraging,
threatening, justifying,
glorifying, or celebrating
physical harm against a group.
* mocking: Cruelly ridiculing or
making fun of a group's
characteristics, identity,
suffering, or stereotypes.
* slurs: Using derogatory,
insulting, or hateful terms
directed at a group.
* exclusion: Suggesting that a
group does not belong, should be
removed, deported, segregated,
or excluded from society.
* not_hateful: The meme does not
contain any of the above
categories.
Multiple labels may apply
simultaneously.
If none of the hateful categories
apply, return only: ["
not_hateful"]
The label "not_hateful" is mutually
exclusive with all other labels
. Never combine "not_hateful"

```

```

with another label.
YOUR OUTPUT MUST BE ONLY A JSON
ARRAY OF STRINGS containing all
applicable labels.
Do not provide explanations,
reasoning, markdown, or any
additional text.

```

A.3 Final MAMI Testing Prompts

Two variants were tested for the MAMI dataset: a standard categorical prompt and a research-grounded prompt.

Listing 5: Custom Categorical Prompt (MAMI Final Test)

```

Act as an expert Trust and Safety
content moderator.
analyse both the image and the
accompanying meme text. The text
extracted from the meme is: "{
clean_text}"
Consider the meaning created by the
combination of the image and
the text. Do not classify based
on the image or text alone.
Assign one or more of the following
labels according to these
definitions:
* stereotype: Portraying women
using harmful generalizations,
fixed negative roles, or
assumptions about what women are
like or should do.
* shaming: Humiliating, degrading,
or ridiculing women for their
appearance, body, behaviour,
sexuality, age, or personal
choices.
* objectification: Reducing women
to sexual objects, body parts,
appearances, commodities, or
possessions rather than treating
them as people.
* violence: Encouraging,
threatening, justifying,
glorifying, or joking about
physical, sexual, or
psychological harm against women
.
* not_misogynistic: The meme does
not contain any of the above
categories.
Multiple labels may apply
simultaneously.
STRICT OUTPUT RULES:

```

```

- If the meme is misogynistic,
return ONLY labels from: ["
stereotype", "shaming", "
objectification", "violence"].
One or more may apply.
- If the meme is NOT misogynistic,
return exactly: ["
not_misogynistic"]
- NEVER mix "not_misogynistic" with
any other label in the same
array.
- NEVER use the word "misogynistic"
as a label.
- YOUR OUTPUT MUST BE ONLY A JSON
ARRAY OF STRINGS.
- Do not provide explanations,
reasoning, markdown, or any
additional text.

```

Listing 6: Research-Based Prompt (MAMI Variant)

```

Act as an expert Trust and Safety
content moderator.
analyse both the image and the
accompanying meme text. The text
extracted from the meme is: "{
clean_text}"
Using generally accepted
definitions from research on
misogyny, gender-based hate, and
online abuse, determine whether
the meme contains misogynistic,
degrading, exclusionary, or
violence-promoting content
directed at women.
Assign one or more of the following
labels: [stereotype, shaming,
objectification, violence,
not_misogynistic]
Consider the meaning created by the
combination of the image and
the text. Do not classify based
on the image or text alone.
Multiple labels may apply
simultaneously. If none of the
misogynistic categories apply,
return only: ["not_misogynistic
"]
YOUR OUTPUT MUST BE ONLY A JSON
ARRAY OF STRINGS containing all
applicable labels.
Do not provide explanations,
reasoning, markdown, or any
additional text.

```

A.4 Experimental European Union (EU) Variants

These prompts were used to compare standard moderation performance against principles derived from the EU Code of Conduct on Countering Illegal Hate Speech Online.

Listing 7: EU Principles Prompt (FHM-FG Comparison)

```
Act as an expert Trust and Safety
content moderator.
analyse both the image and the
accompanying meme text. The text
extracted from the meme is: "{
clean_text}"
Using the principles of the
European Union Code of Conduct
on Countering Illegal Hate
Speech Online, determine whether
the meme contains hateful,
discriminatory, degrading,
exclusionary, or violence-
promoting content directed at a
group.
Assign one or more of the following
labels: [contempt, inferiority,
dehumanising, inciting_violence
, mocking, slurs, exclusion,
not_hateful]
Consider the meaning created by the
combination of the image and
the text. Do not classify based
on the image or text alone.
Multiple labels may apply
simultaneously. If none of the
hateful categories apply, return
only: ["not_hateful"]
YOUR OUTPUT MUST BE ONLY A JSON
ARRAY OF STRINGS containing all
applicable labels.
Do not provide explanations,
reasoning, markdown, or any
additional text.
```

```
- Refer to specific content from
the meme.
- Explain what the meme is saying
or depicting.
- Explain why that content supports
the assigned label(s).
- Use evidence from the meme itself
.
- Not simply repeat the label
definition.
- Be exactly one sentence.
Output exactly in this format:
Labels: ["label1", "label2"]
Reason: one sentence
Do not output any additional text.
```

Listing 8: EU Principles with Single-Sentence Explanation

```
Act as an expert Trust and Safety
content moderator. [Intro
truncated, same as above]...
If "not_hateful" applies, it must
be the only label returned.
After assigning the labels, provide
a single sentence explaining
the decision.
The explanation must:
```